Brevan Ellefsen

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Research Interests

3-manifolds; essential and measured laminations; essential surfaces; foliations; normal surface theory; pseudo-Anosov flows and dynamics; knot theory.

Education

University of Illinois Urbana-Champaign — Champaign, Illinois
Ph.D. in Mathematics, expected May 2027; advisor: Nathan Dunfield — 2021–present

Brigham Young University — Provo, Utah
B.S. in Mathematics — 2017-2020

Research Appointment

SLMath Program Associate — Berkeley, California
Topological and Geometric Structures in Low Dimensions — March 16–April 24, 2026

Publications and Preprints

1. **Brevan Ellefsen**. *Growth Rate of Essential Surfaces via Measured Laminations*. 2026. arXiv:2608.08833 [math.GT].

Research in Progress

- **Least-Weight Laminations.** Developing a normal-surface theory of least-weight representatives of essential measured laminations.
- **Volume Bounds on Growth.** Investigating bounds on measured-lamination-space dimension, and hence essential-surface growth, in terms of hyperbolic volume.
- **Minimal Laminations.** Studying when essential laminations admit representatives whose leaves are minimal for a suitable ambient metric.

Invited Research Talks

- **Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS)** — Online (two-part invited seminar)
  Growth Rate of Essential Surfaces via Measured Laminations — October 2026 (upcoming)
- **University of Wisconsin–Madison** — Madison, Wisconsin
  Growth Rate of Essential Surfaces via Measured Laminations — September 16, 2026 (upcoming)
- **AMS Fall Central Sectional Meeting – Special Session on Foliations and 3-Manifolds** — St. Louis, Missouri
  Growth Rate of Essential Surfaces via Measured Laminations — October 18–19, 2025

Contributed Research Talks

- **Topological and Geometric Structures in Low Dimensions** — SLMath
  Growth Rate of Essential Surfaces via Measured Laminations — February 5, 2026

- **Topological and Geometric Structures in Low Dimensions (Summer Workshop)** — SLMath, July 28, 2025
  Growth Rate of Essential Surfaces via Measured Laminations
- **42nd Workshop in Geometric Topology** — Oberlin College, June 12–14, 2025
  Growth Rate of Essential Surfaces via Measured Laminations
- **Log Cabin Workshop on Foliations** — St. George, Utah, January 13–17, 2025
  Universal Circles, Part 3 (expository talk)

Selected Conferences

Computational Problems about 3-Manifolds, Associated Groups and Varieties — Rutgers University, October 4–5, 2025
Participant

The St. Louis Topology Conference: Flows and Foliations in 3-Manifolds — St. Louis, Missouri, May 17–19, 2024
Participant

Honors and Funding

Departmental Fellowship, University of Illinois Urbana-Champaign
Graduate fellowship support — Spring 2027

Departmental Fellowship, University of Illinois Urbana-Champaign
Graduate fellowship support — Fall 2025–Summer 2026

Teaching Experience

University of Illinois Urbana-Champaign
Teaching Assistant: Calculus I (4 semesters), Calculus II (4 semesters), Linear Algebra (2 semesters) — 2021–present

Brigham Young University
Teaching Assistant: Complex Analysis (1 semester) — 2020

Mentoring and Service

Undergraduate Mentor, UIUC Directed Reading Program — University of Illinois Urbana-Champaign, 3 semesters
Mentored undergraduates in dynamics and algebraic topology

Organizer, UIUC Geometric Topology Seminar — University of Illinois Urbana-Champaign, 2025–present
Created and organize an ongoing seminar for junior researchers in geometric topology and geometric group theory